INTEGRATION TEST PLAN

Airbag Electronic Control Unit (ECU) Portfolio

Measurement and Processing of Sensor Signals to Control Airbag Operation

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Contents

1. Purpose and Scope

This Integration Test Plan (ITP) defines the test strategy, test environment, entry and exit criteria, test case catalogue, and requirement traceability for verifying the interfaces and interactions between the Airbag ECU software modules, between the two redundant Arduino Due (SAM3X8E) channels, and between the firmware and the surrounding hardware (sensors, AND-gate enforcement layer, CAN bus, and the OMS laboratory PC).

Integration testing sits between unit testing and validation testing in the V&V lifecycle. Unit-level correctness of individual modules (MOD-01 through MOD-04) is established separately in the Unit Test Plan, and system-level demonstration of the safety goals (SG-01 through SG-07) is established separately in the Validation Test Plan. This plan assumes unit testing has been completed successfully and focuses exclusively on:

- Module-to-module data flow on the assembled firmware — e.g., sensor acquisition feeding crash detection, crash detection feeding deployment control, diagnostic monitor feeding deployment control and the DWI output.
- Inter-channel (ECU1 ↔ ECU2) interaction over the shared CAN bus, including heartbeat exchange, status frame parsing, and communication-loss handling.
- Hardware/software interaction — ADC sampling, digital I/O, the DM74LS08N AND-gate 2oo2 enforcement layer, indicator LEDs, and the CAN transceiver interface to the OMS laboratory PC.
- Timing behaviour of the integrated system — loop cycle time, crash detection latency, CAN heartbeat interval, communication-loss detection, and DWI activation latency.
- The 12-step HW/SW Integration Bring-Up procedure that must be completed before any of the above interface tests are executed.

The plan covers the integration test cases IT-01 through IT-15 as specified in SADS-AIRBAG-ECU-001 Rev 2.0 Section 9, and the HW/SW integration bring-up procedure specified in SADS-AIRBAG-ECU-001 Rev 2.0 Section 9.1. Unit test cases and validation test cases are out of scope for this document. SADS-AIRBAG-ECU-001 Rev 2.0 records unit, integration, and validation testing, together with the code review checklist, as the responsibility of the Implementation Test Plan (ITP-AIRBAG-ECU-001) and the Test Report (TR-AIRBAG-ECU-001). This document, ITP-AIRBAG-ECU-003 (Integration Test Plan), is a focused companion document that extracts and elaborates only the integration-level test cases and bring-up procedure relevant to module-to-module, inter-channel, and HW/SW interface verification; it does not replace ITP-AIRBAG-ECU-001.

This plan must be read in conjunction with:

- SADS-AIRBAG-ECU-001 Rev 2.0 — Software Architecture Design Specification and Implementation
- SRS-AIRBAG-SW-001 Rev 2.1 — Software Requirements Specification
- FSRS-AIRBAG-ECU-001 Rev 2.1 — Functional Safety Requirements Specification
- HADS-AIRBAG-ECU-001 Rev 2.1 — Hardware Architecture and Design Specification
- ITP-AIRBAG-ECU-001 Rev 1.0 — Implementation Test Plan (unit, integration, and validation test cases; code review checklist)

NOTE: Eight open compliance gaps (GAP-01 through GAP-08) exist in the current firmware, as documented in SADS-AIRBAG-ECU-001 Rev 2.0 Section 10. Section 10 of this plan documents which of those gaps affect the integration test cases in scope here, with a test-execution disposition for each. Gaps relevant only to unit-level or validation-level testing are noted but not dispositioned in this document.

2. Test Strategy

2.1 Position in the V&V Lifecycle

Test Level	Scope	Document
Unit Testing	Each firmware module (MOD-01 to MOD-04) verified in isolation on target hardware.	ITP-AIRBAG-ECU-001 (Implementation Test Plan)
Integration Testing	Module-to-module, inter-channel (ECU1/ECU2), and HW/SW interface verification on the assembled system, plus the bring-up procedure.	This document (ITP-AIRBAG-ECU-003)
Validation Testing	Safety goal (SG-01 to SG-07) demonstration at complete-system level.	ITP-AIRBAG-ECU-001 (Implementation Test Plan)

2.2 Integration Test Level Overview

SADS-AIRBAG-ECU-001 Rev 2.0 Section 9 (HW/SW Integration Procedure) defines the 12-step bring-up procedure referenced in Section 7 of this plan. The integration test cases IT-01 through IT-15 below elaborate the module-to-module and inter-channel interfaces implied by the architecture in SADS Sections 4 through 8 (module decomposition, data flow, state machine, and safety mechanisms) and are grouped as follows:

Sub-Level	Focus	SADS Reference	Test Case Range
HW/SW Integration Bring-Up	Flash, pin, CAN bus, and AND-gate verification — prerequisite for all interface tests below.	Section 9 / 9.1	Bring-up Steps 1–12
Normal Operation Interface Tests	Sensor → crash detection → deployment → AND gate data flow; ECU-to-ECU CAN heartbeat.	Sections 4.2, 6.2, 6.3, 6.5	IT-01 to IT-05
Fault Injection Interface Tests	Diagnostic monitor → deployment control interaction; CAN link-loss handling between channels; AND-gate single-channel rejection.	Sections 6.6, 8.2, 8.3	IT-06 to IT-10
Timing / Interface Latency Tests	Cross-module and cross-channel timing of the integrated system.	Sections 4.3, 5.2	IT-11 to IT-15

2.3 Test Approach

All tests are executed on target hardware (two Arduino Due SAM3X8E boards) to avoid simulator discrepancies, per SADS Section 8.1. Black-box (BB) tests drive inputs (potentiometers, baby seat switch, CAN bus connection) and observe outputs (LEDs, CAN frames, Serial diagnostics) without reference to internal code structure. White-box (WB) tests additionally inspect internal state via Serial Monitor logs (e.g., fault flags, voteDeploy, remoteCrash) to confirm the correct internal interface values are being exchanged between modules and between channels.

The 2oo2 dual-channel architecture is the central interface under test: both ECU boards are operated simultaneously, exchanging status over the shared CAN bus, and the hardware AND gate enforces physical agreement between their D8/D9 outputs. The integration tests explicitly verify that the CAN interface correctly carries crash, fault, and baby-seat status between channels, and that a single-channel fault or single-channel crash detection cannot, by itself, fire any airbag LED.

3. Test Environment

3.1 Hardware Configuration

Component	Specification	Role
Arduino Due x 2	SAM3X8E ARM Cortex-M3, 84 MHz	ECU 1 (ID=1, TX=0x100) and ECU 2 (ID=2, TX=0x200)
POTI-1 (Accelerometer)	10 kΩ potentiometer on A0, 0–3.3 V → 0–100 g	Crash sensor simulation
POTI-2 (Gyroscope)	10 kΩ potentiometer on A1, 0–3.3 V → 0–360 °/s	Rollover sensor simulation
POTI-3 (Pressure)	10 kΩ potentiometer on A2, 0–3.3 V → 0–500 kPa	Side-impact sensor simulation
Baby Seat Switch	Momentary push switch on D11, 10 kΩ pull-up to 3.3 V, active-LOW	Occupant classification
DM74LS08N AND Gate	Quad 2-input AND, pins 1/2 → Airbag1 out, pins 4/5 → Airbag2 out	Hardware 2oo2 enforcement
Airbag1 LED (Red)	3 mm RED LED + 220 Ω from AND gate pin 3	Driver airbag deployment indicator
Airbag2 LED (Red)	3 mm RED LED + 220 Ω from AND gate pin 6	Passenger airbag deployment indicator
DWI LED (Red)	3 mm LED + 220 Ω shared between D10 of ECU1 and ECU2	Diagnostic Warning Indicator
Status LED (D13)	Built-in Arduino LED	Heartbeat / status blink
Waveshare SN65HVD230 x 2	CAN transceiver 3.3 V compatible	CAN bus interface for each ECU
USB-CAN-A adapter	500 kbit/s CAN to USB bridge	OMS PC CAN monitoring interface

3.2 Software / Tooling

Tool	Version	Role
Arduino IDE	2.x	Firmware compilation and upload
GCC ARM (avr-g++)	4.8.3 (bundled)	Cross-compiler for SAM3X8E
due_can library	v1.0 (open source)	CAN0 hardware abstraction
USBCANV2 (OMS PC)	Latest stable	Passive CAN frame monitoring at 500 kbit/s
Serial Monitor	Bundled in Arduino IDE	115200 baud diagnostic/Serial log capture
Git	Any	Firmware version control and release tagging

3.3 Interfaces Under Test

Both ECU boards share the same POTI-1/2/3 analog lines and the same baby seat switch (D11) as defined in HADS-AIRBAG-ECU-001 Rev 2.1. The following table summarises the discrete interfaces exercised by the integration test cases in Section 6, each of which corresponds to a data-flow link described in SADS Section 3.2.

Interface	Endpoints	Carried Data	Exercised By
Sensor → Software (ADC)	POTI-1/2/3, D11 → MOD-02 (Sensor Acquisition)	rawA0, rawA1, rawA2, babySeatRaw	IT-01 to IT-04, IT-12
MOD-02 → MOD-03	Sensor Acquisition → Crash Detection	Scaled accel, gyro, press; localCrash	IT-01 to IT-04, IT-06, IT-12

Interface	Endpoints	Carried Data	Exercised By
MOD-03/MOD-06 → MOD-04	Crash Detection + Diagnostic Monitor → Deployment Control	localCrash, systemFault, babySeat	IT-01 to IT-04, IT-06, IT-09, IT-10
MOD-05 (ECU1) ↔ MOD-05 (ECU2)	CAN0 TX/RX, shared CANH/CANL bus	remoteCrash, remoteFault, lastRemoteHB (status byte bits 0/1/2)	IT-05 to IT-09, IT-13, IT-14
MOD-04 → AND Gate / LEDs	D8, D9, D10 outputs → DM74LS08N → Airbag1/Airbag2/DWI LEDs	voteDeploy, voteDeploy && !babySeat, systemFault	IT-01 to IT-04, IT-06, IT-09, IT-10, IT-15
MOD-05 → OMS PC	CAN0 TX → USB-CAN-A → USBCANV2	8-byte status frame (rawA0/A1/A2, status byte, ECU_ID)	IT-05, IT-13

3.4 Baseline Configuration

Unless an individual test case specifies otherwise, the following baseline configuration applies for all integration tests in Section 6 (per SADS §9.1):

- Both ECUs powered, flashed with the build under test, and communicating (faultCOMM = 0 on both).
- All three potentiometers (POTI-1/2/3) set below their deployment thresholds (ADC < 1000) unless the test requires a crash condition.
- Baby seat switch open (D11 = HIGH, babySeat = FALSE).
- Serial Monitor open at 115200 baud on both ECUs; USBCANV2 running and logging CAN IDs 0x100 and 0x200.
- The shared CAN bus is intact and the USB-CAN-A adapter is in passive monitoring mode — it must not inject frames unless a test case specifies otherwise.

4. Entry and Exit Criteria

4.1 Entry Criteria

The following conditions must be satisfied before integration test execution begins:

#	Entry Criterion	Evidence Required
EC-01	Firmware build corresponding to SADS-AIRBAG-ECU-001 Rev 2.0 complete and version-tagged in Git for both ECUs.	Git tags present for ECU1 and ECU2; confirmed in startup Serial banner (ECU 1 / ECU 2 STARTED).
EC-02	Unit testing (per ITP-AIRBAG-ECU-001) completed with all results recorded.	ITP-AIRBAG-ECU-001 execution record / sign-off reference.
EC-03	Code review checklist (per ITP-AIRBAG-ECU-001, referenced from SADS §11.1) completed with all items PASS.	Signed code review form, two-person review.
EC-04	Test hardware assembled per HADS-AIRBAG-ECU-001 Rev 2.1 wiring tables; continuity check passed.	Continuity check sign-off.
EC-05	Known gaps GAP-01 through GAP-08 (SADS §10) reviewed for relevance to integration test cases.	Gap disposition table in Section 10 of this ITP updated.

4.2 HW/SW Integration Bring-Up as Gating Step

The 12-step HW/SW Integration Bring-Up procedure (Section 7 of this plan) is itself part of integration testing, but it additionally serves as an entry gate for the interface test cases IT-01 through IT-15: those tests shall not commence until all 12 bring-up steps have a recorded PASS and the Integration Sign-Off in Section 7 has been completed.

4.3 Exit Criteria

#	Exit Criterion
EX-01	All 12 HW/SW Integration Bring-Up steps executed; results recorded as PASS with sign-off completed.
EX-02	All integration tests IT-01 through IT-15 executed; results recorded as PASS, FAIL, or WAIVED with justification.
EX-03	Integration-level coverage targets in Section 8 met, or a documented shortfall and rationale recorded.
EX-04	All FAIL results have an approved deviation report or corrective action plan.
EX-05	Test evidence archived per Section 9: Serial logs, USBCANV2 logs, LED photos/video, operator/date/firmware version for every executed test case.
EX-06	Integration-relevant gap dispositions in Section 11 reviewed and accepted.
EX-07	V&V lead and functional safety engineer sign-off obtained (Section 12).

Completion of this plan's exit criteria is a precondition for the Validation Test Plan (VT-01 to VT-07) to commence, since validation testing assumes that the integrated system's module-to-module, channel-to-channel, and HW/SW interfaces have already been verified.

5. Integration Test Case Format

Each integration test case below is documented using a common field structure: Test ID, Test Type (BB = black-box, WB = white-box, BB+WB = combined), Test Name, the interface(s) under test, FSR trace, ASIL, preconditions, test procedure, expected result, pass criterion, evidence required, and a Result/Notes field to be completed during execution. The 'Interface Under Test' field, specific to this Integration Test Plan, identifies which data-flow link from Section 3.3 is being exercised.

6. Integration Test Cases

6.1 Normal Operation Tests

These tests verify the primary data-flow path — sensors through crash detection and deployment control to the AND-gate outputs — and the basic ECU1↔ECU2 CAN heartbeat interface, under fault-free conditions.

IT-01 — No-crash baseline — all sensors below threshold

Test ID	IT-01
Test Type	BB (Black-Box)
Test Name	No-crash baseline — all sensors below threshold
Interface Under Test	Sensor → MOD-02 → MOD-03 → MOD-04 → AND Gate / LEDs (Section 3.3)
FSR Trace	FSR-04
ASIL	ASIL D
Preconditions	Full assembled system: both ECUs, CAN bus, AND gate, LEDs. USBCANV2 running. Serial Monitor open on both ECUs. Baseline configuration per Section 3.4.
Test Procedure	Both ECUs powered and communicating. Set all POTIs well below threshold (ADC < 1000 on all channels). Baby seat switch OPEN. Run for minimum 5 minutes.

Expected Result	Both Airbag LEDs remain OFF for entire 5-minute run. DWI OFF. Serial: localCrash=0, voteDeploy=0 on both ECUs continuously.
Pass Criterion	Zero false deployments; Airbag LEDs OFF for ≥ 5 minutes without interruption.
Evidence Required	Serial logs from both ECUs (timestamped). USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-02 — Full dual-channel crash — both ECUs above threshold simultaneously

Test ID	IT-02
Test Type	BB (Black-Box)
Test Name	Full dual-channel crash — both ECUs above threshold simultaneously
Interface Under Test	MOD-03 → MOD-04 → AND Gate (2oo2 agreement); MOD-05 ECU1 ↔ ECU2 status exchange
FSR Trace	FSR-01, FSR-03
ASIL	ASIL D
Preconditions	Full assembled system: both ECUs, CAN bus, AND gate, LEDs. USBCANV2 running. Serial Monitor open on both ECUs. Baseline configuration per Section 3.4.
Test Procedure	Both ECUs communicating. Simultaneously advance POTI-1 above threshold on both ECUs (ADC > 1228). Observe LEDs and Serial on both ECUs. Monitor USBCANV2 for crash bit in frames 0x100 and 0x200.
Expected Result	Airbag1 LED fires ON. Serial: voteDeploy=1 on both ECUs. USBCANV2 shows bit 0 of byte 6 = 1 in both 0x100 and 0x200 frames.
Pass Criterion	Airbag1 LED ON within observed cycle; USBCANV2 confirms crash bit in both CAN frames.
Evidence Required	Serial logs from both ECUs (timestamped). USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-03 — Crash ceases — sensors returned below threshold

Test ID	IT-03
Test Type	BB (Black-Box)
Test Name	Crash ceases — sensors returned below threshold
Interface Under Test	MOD-03 → MOD-04 → AND Gate / LEDs — verifies the interface returns to its rest state without latching
FSR Trace	FSR-04
ASIL	ASIL D
Preconditions	Full assembled system: both ECUs, CAN bus, AND gate, LEDs. USBCANV2 running. Serial Monitor open on both ECUs. Baseline configuration per Section 3.4.
Test Procedure	Following IT-02, return POTI-1 below threshold on both ECUs.
Expected Result	Airbag1 and Airbag2 LEDs extinguish within one loop cycle of POTI-1 dropping below threshold. Serial: localCrash=0, voteDeploy=0.
Pass Criterion	LEDs extinguish within one cycle of sensor drop below threshold.

Evidence Required	Serial logs from both ECUs (timestamped). USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-04 — Baby seat suppression during crash

Test ID	IT-04
Test Type	BB (Black-Box)
Test Name	Baby seat suppression during crash
Interface Under Test	Baby seat switch (D11) → MOD-02 → MOD-04 → D9 (Airbag2) — verifies the babySeat interface independently suppresses only the passenger channel
FSR Trace	FSR-09
ASIL	ASIL D
Preconditions	Full assembled system: both ECUs, CAN bus, AND gate, LEDs. USBCANV2 running. Serial Monitor open on both ECUs. Baseline configuration per Section 3.4.
Test Procedure	Both ECUs above threshold (crash condition). Press and hold baby seat switch (D11 LOW). Observe D8 and D9.
Expected Result	Airbag1 LED ON (D8 HIGH on both ECUs). Airbag2 LED OFF (D9 LOW on both ECUs). Serial: babySeat=1.
Pass Criterion	Airbag2 LED stays OFF; Airbag1 LED remains ON during baby seat + crash condition.
Evidence Required	Serial logs from both ECUs (timestamped). USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-05 — CAN heartbeat visible on USBCANV2

Test ID	IT-05
Test Type	BB (Black-Box)
Test Name	CAN heartbeat visible on USBCANV2
Interface Under Test	MOD-05 → OMS PC (CAN0 TX → USB-CAN-A → USBCANV2) — verifies the ECU-to-OMS monitoring interface
FSR Trace	FSR-16
ASIL	ASIL D
Preconditions	Full assembled system: both ECUs, CAN bus, AND gate, LEDs. USBCANV2 running. Serial Monitor open on both ECUs. Baseline configuration per Section 3.4.
Test Procedure	Both ECUs powered and communicating normally. Open USBCANV2 at 500 kbit/s. Record frame timestamps for CAN IDs 0x100 and 0x200 over 20 consecutive frames each.
Expected Result	Frames for both 0x100 and 0x200 arrive at ~500 ms intervals. All inter-frame gaps within 450–550 ms.
Pass Criterion	All 20 intervals for both CAN IDs within 450–550 ms.
Evidence Required	Serial logs from both ECUs (timestamped). USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version.
Result (Pass/Fail)	

Notes	

6.2 Fault Injection Tests

These tests verify the diagnostic-monitor-to-deployment-control interface, the inter-channel fault-status exchange over CAN, and the AND-gate's hardware enforcement of channel agreement when one side of an interface reports a fault or an out-of-agreement crash state.

IT-06 — Single-channel crash — only ECU1 above threshold, ECU2 below

Test ID	IT-06
Test Type	BB
Test Name	Single-channel crash — only ECU1 above threshold, ECU2 below
Interface Under Test	MOD-04 (ECU1) → D8 → AND Gate ← D8 MOD-04 (ECU2) — verifies the AND gate's 2oo2 interface rejects a single-channel assertion; MOD-05 ECU1 ↔ ECU2 remoteCrash exchange
FSR Trace	FSR-03, FSR-08
ASIL	ASIL D
Preconditions	Full assembled system. USBCANV2 running. Serial Monitor on both ECUs. System in ST-03 MONITORING state unless stated.
Test Procedure	Set POTI-1 above threshold on ECU1 only (ADC > 1228). Keep all POTIs on ECU2 below threshold. Observe all LEDs and Serial remoteCrash on ECU1.
Expected Result	Both Airbag LEDs OFF. ECU1 Serial: localCrash=1, remoteCrash=0 (ECU2 not detecting crash), voteDeploy=0. AND gate cannot assert because ECU2 D8 is LOW.
Pass Criterion	No airbag LED fires when only one ECU detects a crash; AND gate hardware prevents single-channel deployment.
Evidence Required	Serial logs from both ECUs. USBCANV2 log. LED photograph. Timing measurement where applicable. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-07 — CAN bus disconnected — peer timeout after 2000 ms

Test ID	IT-07
Test Type	BB
Test Name	CAN bus disconnected — peer timeout after 2000 ms
Interface Under Test	MOD-05 ECU1 ↔ ECU2 CAN link (heartbeat/timeout) → MOD-06 (faultCOMM) → MOD-04 / DWI — verifies loss of the inter-channel interface is correctly propagated to the deployment and warning outputs
FSR Trace	FSR-15, FSR-20
ASIL	ASIL D
Preconditions	Full assembled system. USBCANV2 running. Serial Monitor on both ECUs. System in ST-03 MONITORING state unless stated.
Test Procedure	Both ECUs in normal monitoring state (POTIs below threshold). Physically disconnect the CAN bus (remove CANH or CANL wire). Start timing from disconnection. Observe DWI LED and Serial faultCOMM flag.
Expected Result	After 2000 ms without a valid peer CAN frame: faultCOMM=TRUE on both ECUs, DWI LED ON, D8=LOW, D9=LOW. Serial shows systemFault=1.
Pass Criterion	DWI ON within 2000 ms of CAN disconnection; airbag LEDs remain OFF.

Evidence Required	Serial logs from both ECUs. USBCANV2 log. LED photograph. Timing measurement where applicable. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-08 — CAN reconnected after timeout — system recovers to MONITORING

Test ID	IT-08
Test Type	BB+WB
Test Name	CAN reconnected after timeout — system recovers to MONITORING
Interface Under Test	MOD-05 ECU1 ↔ ECU2 CAN link — verifies the inter-channel interface re-establishes cleanly and faultCOMM clears without manual intervention
FSR Trace	FSR-15
ASIL	ASIL D
Preconditions	Full assembled system. USBCANV2 running. Serial Monitor on both ECUs. System in ST-03 MONITORING state unless stated.
Test Procedure	Following IT-07, reconnect CAN bus. Observe faultCOMM and DWI LED after next valid peer CAN frame is received.
Expected Result	faultCOMM clears to FALSE within one CAN frame period (~500 ms). DWI LED OFF. System returns to ST-03 MONITORING. Serial: faultCOMM=0.
Pass Criterion	System recovery within one CAN heartbeat period after CAN reconnection; DWI extinguishes.
Evidence Required	Serial logs from both ECUs. USBCANV2 log. LED photograph. Timing measurement where applicable. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-09 — Remote fault bit set — peer broadcasts fault in CAN byte 6

Test ID	IT-09
Test Type	WB
Test Name	Remote fault bit set — peer broadcasts fault in CAN byte 6
Interface Under Test	MOD-05 (ECU2) → CAN status byte bit 2 → MOD-05 (ECU1) remoteFault → MOD-04 — verifies the remote-fault field of the CAN status interface correctly inhibits deployment on the receiving channel
FSR Trace	FSR-03
ASIL	ASIL D
Preconditions	Full assembled system. USBCANV2 running. Serial Monitor on both ECUs. System in ST-03 MONITORING state unless stated.
Test Procedure	Force ECU2 to set bit 2 of its CAN status byte (systemFault=TRUE broadcast). Simultaneously set all sensors above threshold on both ECUs. Observe voteDeploy on ECU1.
Expected Result	ECU1: remoteFault=TRUE received; voteDeploy=0; D8=LOW, D9=LOW even with both sensors above threshold. Airbag LEDs OFF.
Pass Criterion	Deployment inhibited when peer ECU reports fault, regardless of sensor state.
Evidence Required	Serial logs from both ECUs. USBCANV2 log. LED photograph. Timing measurement where applicable. Operator name, date, firmware version.

Result (Pass/Fail)	
Notes	

IT-10 — POST failure simulation — faultCPU forced TRUE on ECU1

Test ID	IT-10
Test Type	BB
Test Name	POST failure simulation — faultCPU forced TRUE on ECU1
Interface Under Test	MOD-01 (POST, ECU1) → MOD-06 (systemFault) → MOD-04 → D8 → AND Gate — verifies a startup-fault interface on one channel propagates through to the shared 2oo2 hardware interface and prevents the peer's otherwise-valid crash signal from firing the LED
FSR Trace	FSR-17, FSR-20
ASIL	ASIL D
Preconditions	Full assembled system. USBCANV2 running. Serial Monitor on both ECUs. System in ST-03 MONITORING state unless stated.
Test Procedure	Build ECU1 firmware with cpuTest() returning FALSE. Flash. Power-cycle. Set all POTIs above threshold on both ECUs. Observe for entire session.
Expected Result	DWI ON from startup. voteDeploy=0 on ECU1. D8 on ECU1 remains LOW. Since AND gate requires both D8 HIGH simultaneously, Airbag1 LED stays OFF even with ECU2 in full crash state.
Pass Criterion	No airbag LED fires throughout entire session when ECU1 has POST fault; DWI ON from startup.
Evidence Required	Serial logs from both ECUs. USBCANV2 log. LED photograph. Timing measurement where applicable. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

6.3 Timing / Interface Latency Tests

These tests measure the timing behaviour of the integrated cross-module and cross-channel interfaces — the loop cycle that ties MOD-02 through MOD-06 together, the sensor-to-output latency through the crash-detection-to-deployment interface, the CAN heartbeat interface timing, the CAN-loss detection latency, and the fault-to-DWI interface latency.

IT-11 — Loop cycle time ≤ 1 ms

Test ID	IT-11
Test Name	Loop cycle time ≤ 1 ms
Interface Under Test	Cyclic executive — aggregate timing of the MOD-02 → MOD-03 → MOD-06 → MOD-04 → MOD-05 interface chain within one loop() iteration
FSR Trace	FSR-02
ASIL	ASIL D
Preconditions	Full assembled system in MONITORING state. Timing-instrumented firmware build where stated.
Test Procedure	Add micros() timestamps at start and end of loop() in a timing-instrumented build. Log via Serial over 10,000 consecutive iterations.
Measurement Method	Record maximum, mean, and 99th percentile cycle time from the 10,000-sample log.

Pass Criterion	99th percentile cycle time ≤ 1000 µs (1 ms).
Evidence Required	Serial timing log with sample count. USBCANV2 timestamp log. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-12 — Crash detection response ≤ 20 ms

Test ID	IT-12
Test Name	Crash detection response ≤ 20 ms
Interface Under Test	Sensor (POTI-1) → MOD-02 → MOD-03 (localCrash) interface latency
FSR Trace	FSR-01
ASIL	ASIL D
Preconditions	Full assembled system in MONITORING state. Timing-instrumented firmware build where stated.
Test Procedure	Log millis() immediately before POTI-1 is moved above threshold and immediately when localCrash=TRUE appears in Serial output. Repeat 10 trials.
Measurement Method	Delta between threshold crossing and localCrash=TRUE logged for each trial.
Pass Criterion	All 10 deltas ≤ 20 ms.
Evidence Required	Serial timing log with sample count. USBCANV2 timestamp log. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-13 — CAN heartbeat interval 500 ms ± 50 ms

Test ID	IT-13
Test Name	CAN heartbeat interval 500 ms ± 50 ms
Interface Under Test	MOD-05 → OMS PC CAN interface — transmit timing of the periodic status frame
FSR Trace	FSR-16
ASIL	ASIL D
Preconditions	Full assembled system in MONITORING state. Timing-instrumented firmware build where stated.
Test Procedure	Log USBCANV2 frame timestamps for 20 consecutive frames from each ECU (0x100 and 0x200).
Measurement Method	Calculate inter-frame intervals from USBCANV2 timestamps.
Pass Criterion	All intervals within 450–550 ms for both CAN IDs.
Evidence Required	Serial timing log with sample count. USBCANV2 timestamp log. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-14 — faultCOMM declared ≤ 2000 ms after CAN loss

Test ID	IT-14

Test Name	faultCOMM declared ≤ 2000 ms after CAN loss
Interface Under Test	MOD-05 ECU1 ↔ ECU2 CAN link — timeout-detection latency of the inter-channel heartbeat interface
FSR Trace	FSR-15
ASIL	ASIL D
Preconditions	Full assembled system in MONITORING state. Timing-instrumented firmware build where stated.
Test Procedure	Log millis() at last received valid CAN frame before disconnecting CAN. Log millis() when faultCOMM=TRUE appears in Serial. Calculate delta.
Measurement Method	Delta = time from last CAN frame to faultCOMM=TRUE in Serial log.
Pass Criterion	faultCOMM declared ≤ 2050 ms after last valid peer frame (NOTE: GAP-08 — FSR-15 requires ≤50 ms; current implementation uses 2000 ms; documented deviation, see Section 10).
Evidence Required	Serial timing log with sample count. USBCANV2 timestamp log. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

IT-15 — DWI activates within 800 ms of fault

Test ID	IT-15
Test Name	DWI activates within 800 ms of fault
Interface Under Test	MOD-06 (fault flags) → MOD-04 → D10 (DWI) — latency of the fault-to-warning-indicator interface
FSR Trace	FSR-19
ASIL	ASIL D
Preconditions	Full assembled system in MONITORING state. Timing-instrumented firmware build where stated.
Test Procedure	Inject fault at known millis() timestamp (disconnect CAN or force faultCPU). Log millis() at D10=HIGH transition in Serial. Calculate delta.
Measurement Method	Delta from fault injection to D10 HIGH in Serial log.
Pass Criterion	D10 HIGH within 800 ms of fault; per design D10 fires within ~1 ms (one cycle), well within FSR-19 limit.
Evidence Required	Serial timing log with sample count. USBCANV2 timestamp log. Operator name, date, firmware version.
Result (Pass/Fail)	
Notes	

7. HW/SW Integration Bring-Up Procedure

The 12-step bring-up procedure from SADS-AIRBAG-ECU-001 Rev 2.0 Section 9.1 verifies the physical and logical interfaces — power rails, firmware flashing, CAN bus bring-up, and the AND-gate hardware interface — before the interface test cases in Section 6 are executed. All steps must show PASS, per the entry criteria in Section 4.2.

Step	Action	Acceptance Criterion	Owner	Result
1	Assemble hardware per HADS wiring tables. Verify breadboard wiring, POTIs, baby seat switch, CAN transceivers, AND gate, LEDs.	All wiring verified against HADS Tables 5–14. No short circuits. Continuity on all signal lines.	HW Engineer	
2	Verify power rails: 5 V rail to AND gate VCC; 3.3 V rail to CAN transceivers; GND common to all.	5 V ± 0.25 V at AND gate pin 14. 3.3 V ± 0.15 V at Waveshare VCC. GND continuity confirmed.	HW Engineer	
3	Flash ECU 1 firmware (ECU_ID=1, DATA_ID=0x100, REMOTE_ID=0x200) via Arduino IDE.	Upload completes 'Done uploading'. Serial: 'ECU 1 STARTED' at 115200 baud.	SW Engineer	
4	Flash ECU 2 firmware (ECU_ID=2, DATA_ID=0x200, REMOTE_ID=0x100) via Arduino IDE.	Upload completes. Serial: 'ECU 2 STARTED' at 115200 baud.	SW Engineer	
5	Verify POST results on both ECUs via Serial Monitor.	Serial: faultCPU=0, faultRAM=0, faultADC=0. D13 blinking at ~500 ms on both ECUs.	SW Engineer	
6	Verify CAN bus bring-up — connect OMS PC via USB-CAN-A, open USBCANV2 at 500 kbit/s.	CAN frames with ID 0x100 and 0x200 visible at ~500 ms intervals in USBCANV2 log.	HW/SW Engineer	
7	Verify CAN communication between ECUs — check faultCOMM=0 on both ECUs.	Serial: faultCOMM=0 on both ECUs. Remote crash and fault bits correctly parsed.	SW Engineer	
8	Verify pin output levels at rest — all sensors below threshold.	D8=LOW, D9=LOW, D10=LOW, D13 blinking. Airbag LEDs OFF. DWI OFF.	HW Engineer	
9	Verify AND gate logic — set POTI-1 on ECU1 only above threshold.	Airbag1 LED stays OFF. AND gate pin 3 LOW. Only ECU1 asserting D8.	HW Engineer	
10	Verify full crash scenario — set both ECU POTI-1 above threshold simultaneously.	Airbag1 LED fires (AND gate pin 3 HIGH). Serial: voteDeploy=1 on both ECUs.	HW/SW Engineer	
11	Verify baby seat suppression — press baby seat switch during step 10 crash.	Airbag1 LED ON; Airbag2 LED OFF. D9 LOW on both ECUs.	HW/SW Engineer	
12	Sign off HW/SW integration — record results and release to interface testing (Section 6).	All steps PASS. Integration sign-off form completed.	Project Lead	

Integration Sign-Off:

Role	Name	Signature	Date
HW Engineer			
SW Engineer			
Project Lead			

8. Coverage Targets

Coverage at the integration level is defined in terms of interfaces and cross-module/crosschannel interaction paths exercised, rather than the statement- and branch-level MC/DC coverage applicable to unit testing (which is addressed in the Unit Test Plan). The following targets apply to this Integration Test Plan.

Coverage Type	Target	Method	Test Cases
Data-Flow Interface Coverage	Every interface listed in Section 3.3 exercised by at least one passing integration test case.	Cross-check each row of the Section 3.3 interface table against the 'Exercised By' column.	IT-01 to IT-15
Inter-Channel (2oo2) Coverage	Both the agreement path (both channels assert) and the disagreement path (single channel asserts, or one channel faulted) exercised.	Dedicated agreement/disagreement test pairs.	IT-02/IT-06, IT-09/IT-02, IT-10
CAN Link Coverage	Normal heartbeat, link loss, and link recovery all exercised on the inter-channel CAN interface.	Sequential normal → loss → recovery test sequence.	IT-05, IT-07, IT-08
Timing Interface Coverage	Every timing requirement associated with a cross-module or cross-channel interface (FSR-02, FSR-01, FSR-16, FSR-15, FSR-19) measured at least once.	Instrumented timing builds with Serial/USBCANV2 timestamp logging.	IT-11 to IT-15
HW/SW Interface Coverage	Power, flashing, CAN bus bring-up, pin-level rest state, and AND-gate logic all verified before interface testing begins.	12-step bring-up procedure, all steps PASS.	Bring-up Steps 1–12

8.1 Interface Agreement / Disagreement Pairs

The following table cross-references the dual-channel (2oo2) interface conditions that must each be exercised by at least one integration test case, ensuring both the agreement path and every disagreement path through the AND-gate interface are covered.

2oo2 Interface Condition	Agreement / Disagreement	Covered By
Both channels assert D8 (crash confirmed on both)	Agreement — LED fires	IT-02
Only ECU1 asserts D8 (ECU2 below threshold)	Disagreement — AND gate blocks	IT-06
ECU1 D8 forced LOW by local POST fault, ECU2 in full crash	Disagreement — AND gate blocks	IT-10
ECU1 sensors above threshold but remoteFault=TRUE received from ECU2	Disagreement — AND gate blocks via inhibited vote	IT-09
Both channels report crash, baby seat asserted — D8 agree, D9 individually suppressed	Agreement on D8, independent suppression on D9	IT-04
CAN link lost → faultCOMM on both channels → both D8/D9 forced LOW	Disagreement avoided by mutual inhibit (fail-safe)	IT-07

9. Test Evidence and Reporting

9.1 Required Evidence per Test Case

The following evidence must be collected and archived for every integration test case executed:

Evidence Item	Format	Storage Location
Serial Monitor log (both ECUs)	Timestamped .txt file (115200 baud capture)	Test archive — date-stamped folder per test session
USBCANV2 CAN frame log	Exported .csv or .log from USBCANV2	Test archive — same folder as Serial logs
LED state photograph or video	.jpg / .mp4 capturing Airbag1, Airbag2, DWI LED states at key moments	Test archive — labelled with test case ID
Firmware version record	From startup Serial banner: firmware revision, ECU ID, build date/time (__DATE__, __TIME__)	Included in Serial log and test report header
Test operator record	Operator name, date, hardware revision, pass/fail determination	Test report sign-off sheet

9.2 Test Report Structure

A test report shall be produced at the completion of integration test execution. The report shall include:

- Document title, document number, revision, date, and firmware version tested.
- Summary table of all integration test cases: Test ID, name, result (Pass/Fail), and any deviations.
- HW/SW Integration Bring-Up sign-off record (Section 7) attached or referenced.
- For each FAIL: defect description, root cause (if known), corrective action plan or deviation approval reference.
- Coverage summary against the targets in Section 8.
- Test evidence log listing all archived files with their paths and checksums.
- Sign-off table completed by test executor, reviewer (independent), V&V lead, and functional safety engineer.

10. Compliance Gaps Affecting Integration Test Scope

The following gaps between the current firmware and FSRS requirements have been identified in SADS-AIRBAG-ECU-001 Rev 2.0 Section 10 (Compliance Gaps and Required Improvements). Only gaps that affect an interface exercised by an integration test case (IT-01 to IT-15) or the HW/SW bring-up procedure are dispositioned below. Gaps relevant only to unit-level or system-level (validation) testing — GAP-02, GAP-07 — are noted for completeness but carry no integration-test disposition; their required actions are tracked in ITP-AIRBAG-ECU-001.

Gap ID	FSR	Description	Affected Integration Interface	Test Plan Disposition
GAP-01	FSR-11	Baby seat debounce not implemented — D11 read directly without 10 ms stability filter.	Baby seat switch (D11) → MOD-02 → MOD-04 interface (IT-04)	IT-04 executes as-is. Switch bounce may cause intermittent false suppression during crash vibration. Deviation accepted for lab integration testing; closure required before production. Test marked with deviation flag.
GAP-02	FSR-12	Baby seat line continuity monitoring not implemented — open-circuit stuck-HIGH not detected.	Not directly exercised by an IT-XX case	No integration test case defined for this unimplemented function. Out of scope for this ITP — required action (periodic D11

Gap ID	FSR	Description	Affected Integration Interface	Test Plan Disposition
				continuity test) tracked in ITP-AIRBAG-ECU-001 / SADS §10 (GAP-02).
GAP-03	FSR-13	CRC not included in CAN frame — bytes 6–7 carry status and ECU_ID, not a 16-bit CRC.	MOD-05 → OMS PC CAN interface (IT-05, IT-13)	IT-05 and IT-13 proceed without CRC verification; USBCANV2 log used to verify correct byte content directly. GAP-03 closure required before full SG-07 validation.
GAP-04	FSR-14	Parallel CAN bitstream readback not implemented.	MOD-05 → OMS PC CAN interface	No integration test case defined for this unimplemented feature. Deferred — the AND-gate hardware interface provides the second-layer protection exercised instead (IT-02, IT-06).
GAP-05	FSR-17	CAN communication not verified during POST — Can0.begin() return value not checked.	MOD-01 (POST) → MOD-05 CAN interface (Bring-Up Step 5/6)	Bring-up steps 5 and 6 proceed without an explicit POST-time CAN init check. Functional CAN interface verification is provided by IT-05 / IT-06 after bring-up. Deviation noted.
GAP-06	FSR-22	Hardware WDT not enabled in firmware.	Cyclic executive interface chain (IT-11, IT-14, IT-15)	WDT absence means a software lockup during the timing tests would not be caught by hardware. IT-11, IT-14, and IT-15 proceed as specified. WDT implementation required before validation sign-off; flagged as a residual risk for this integration cycle.
GAP-07	FSR-21	Fault logging to DTC/non-volatile memory not implemented.	Not directly exercised by an IT-XX case	No integration test case for EEPROM/flash DTC storage. Out of scope for this ITP — Serial Monitor logs serve as the fault-evidence interface for all integration tests in Section 6.
GAP-08	FSR-15	CAN timeout threshold is 2000 ms vs FSR-15 requirement of ≤50 ms.	MOD-05 ECU1 ↔ ECU2 CAN link timeout interface (IT-07, IT-14)	IT-07 and IT-14 test at the 2000 ms threshold actually implemented. IT-14 pass criterion states ≤ 2050 ms with an explicit note that FSR-15 requires ≤50 ms. Safety rationale: a 2000 ms timeout does not create a deployment risk since CAN loss causes inhibit (fail-safe direction). Formal deviation justification required at validation level.

11. Integration Requirement Traceability Matrix

This matrix lists the FSRs that are verified, in whole or in part, by an integration test case (IT-01 to IT-15) or the HW/SW Integration Bring-Up procedure, together with the implementing Software Requirement (SWR) from SADS-AIRBAG-ECU-001 Rev 2.0 Section 12. FSRs whose verification relies solely on unit testing or validation testing (per ITP-AIRBAG-ECU-001) are not repeated here. Where an

FSR is only partially covered at integration level (the remainder being covered at unit or validation level), this is noted in the Coverage Note column.

FSR ID	SWR(s)	FSR Title	ASIL	Integration Interface (Section 3.3)	Integration Test Case(s)	Coverage Note
FSR-01	SWR-04, SWR-05	Crash Detection Timing (≤20 ms)	ASIL D	Sensor → MOD-02 → MOD-03	IT-02, IT-12	Threshold logic itself unit-tested; latency verified here.
FSR-02	SWR-11	Algorithm Cycle Time (≤1 ms)	ASIL D	Full cyclic executive interface chain	IT-11	Integration-only requirement — measured on assembled system.
FSR-03	SWR-06, SWR-10, SWR-12	Deployment Authorization Logic	ASIL D	MOD-03/MOD-06 → MOD-04 → AND Gate	IT-02, IT-04, IT-06, IT-09	Vote logic unit-tested; cross-channel agreement verified here.
FSR-04	SWR-06	Prevention of Unintended Deployment	ASIL D	MOD-03 → MOD-04 → AND Gate / LEDs	IT-01, IT-03, IT-06	—
FSR-07	SWR-17, SWR-18, SWR-19	Dual-Channel Redundancy	ASIL D	MOD-05 ECU1 ↔ ECU2	IT-06	Full redundancy demonstration at validation level (ITP-AIRBAG-ECU-001).
FSR-08	SWR-12	Channel Agreement Verification	ASIL D	MOD-04 → D8/D9 → AND Gate	IT-06	—
FSR-09	SWR-07, SWR-13	Passenger Airbag Suppression	ASIL D	Baby seat switch → MOD-02 → MOD-04 → D9	IT-04	Switch-level logic unit-tested; cross-module suppression verified here.
FSR-15	SWR-18	CAN Timeout Detection (≤50 ms)	ASIL B	MOD-05 ECU1 ↔ ECU2 link	IT-07, IT-14	GAP-08 — tested at 2000 ms implemented threshold (see Section 10).
FSR-16	SWR-21, SWR-22	Crash Notification Transmission	ASIL B	MOD-05 → OMS PC	IT-05, IT-13	GAP-03 — frame transmitted without 16-bit CRC; content/timing verified here, full CRC verification per ITP-AIRBAG-ECU-001.
FSR-17	SWR-01, SWR-02	Power-On Self-Test	ASIL C	MOD-01 → MOD-05/MOD-06 interface (Bring-up, IT-10)	Bring-up Step 5, IT-10	POST internal logic unit-tested per ITP-AIRBAG-ECU-001; interface to deployment/CAN verified here. GAP-05 — CAN

FSR ID	SWR(s)	FSR Title	ASIL	Integration Interface (Section 3.3)	Integration Test Case(s)	Coverage Note
						init not checked during POST.
FSR-19	SWR-22, SWR-23	DWI Activation (≤800 ms)	ASIL A	MOD-06 → MOD-04 → D10	IT-15	—
FSR-20	SWR-14	Safe-State Transition	ASIL D	MOD-06 → MOD-04 → D8/D9/D10	IT-07, IT-10	—

Note: SADS-AIRBAG-ECU-001 Rev 2.0 Section 12 lists SWR-22 twice (once for CAN TX, once for DWI on local fault) — both instances are referenced above against FSR-16 and FSR-19 respectively, as they appear in the source traceability matrix.

FSR-10, FSR-11, FSR-12, FSR-13, FSR-14, FSR-18, FSR-21, and FSR-22 are addressed in ITP-AIRBAG-ECU-001 (unit and validation test cases) and are referenced in this document only through the gap dispositions in Section 10 where they affect an integration interface (GAP-01, GAP-03, GAP-04, GAP-05, GAP-06, GAP-08).

12. Roles, Responsibilities, and Sign-Off

12.1 Roles and Responsibilities

Role	Responsibility
Test Executor	Executes integration test cases per procedure; captures evidence; records pass/fail; raises defect reports for failures.
Independent Reviewer	Reviews test results against acceptance criteria; verifies evidence completeness; required for sign-off of all ASIL D integration test cases.
V&V Lead	Approves this plan; reviews coverage against Section 8; signs off the integration test report; confirms exit criteria in Section 4.3 are met.
Functional Safety Engineer	Reviews the traceability matrix in Section 11; confirms interface coverage is adequate input to validation; signs off gap dispositions in Section 10.
HW / SW Engineers	Execute and sign off the HW/SW Integration Bring-Up procedure (Section 7).
Project Supervisor	Final authority for plan approval; approves any deviation reports and residual risk acceptances raised during integration testing.

12.2 Test Plan Approval Sign-Off

Role	Name	Signature	Date
Software Safety Engineer			
Hardware Safety Engineer			
Functional Safety Engineer			
V&V Lead			
Project Supervisor			

12.3 Test Execution Sign-Off

Activity	Executor	Reviewer	V&V Lead	Date / Firmware Rev
HW/SW Integration Bring-Up (Steps 1–12)				

Activity	Executor	Reviewer	V&V Lead	Date / Firmware Rev
Normal Operation Tests (IT-01 to IT-05)				
Fault Injection Tests (IT-06 to IT-10)				
Timing / Interface Latency Tests (IT-11 to IT-15)				

13. Revision History

Rev	Date	Author	Description
1.0	2026-06-13	Atharva Banarase — Safety Engineer	Initial release as ITP-AIRBAG-ECU-003, Integration Test Plan, derived from SADS-AIRBAG-ECU-001 Rev 2.0 (final). Scope restricted to interface and inter-channel integration verification: integration tests IT-01 to IT-15 and the 12-step HW/SW integration bring-up procedure (SADS Rev 2.0 Section 9.1). Unit, integration, validation test cases, and the code review checklist remain the responsibility of ITP-AIRBAG-ECU-001 (Implementation Test Plan) and TR-AIRBAG-ECU-001 (Test Report), per SADS Rev 2.0 Section 2. LED hardware corrected to all-RED 3 mm indicators per HADS. Compliance gap dispositions (Section 10) reflect SADS Rev 2.0 Section 10 — GAP-01 through GAP-08 all open. Traceability matrix (Section 11) updated with SWR references from SADS Rev 2.0 Section 12.